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This workshop provided a practical introduction to image classification using deep learning and PyTorch. The primary objective was to build, train, and evaluate a neural network capable of distinguishing between images of chihuahuas and muffins. Although the task may seem simple, it demonstrated the complete machine learning workflow used in many modern computer vision applications. Throughout the workshop, I learned how image data is prepared for training, how neural networks process information, and how model performance is evaluated using validation data. The techniques used included data preprocessing, image transformations, dataset loading, model training, validation, and visualization of predictions. Working through each stage of the project helped me understand how computer vision systems are developed and how machine learning models learn from image data.

One of the most important concepts I learned was image classification. Image classification is the process of assigning a category label to an image based on its visual content. In this workshop, the model learned to classify images as either a chihuahua or a muffin. I also gained a deeper understanding of neural networks and how they learn patterns from data. Before completing this activity, I understood the basic theory behind machine learning, but seeing a model train and improve its predictions helped me connect those concepts to a real implementation. Another important concept was data preprocessing. Images must be resized, normalized, and converted into tensors before they can be processed by a neural network. The workshop also introduced the concepts of convolutional neural networks (CNNs) and transfer learning. CNNs are particularly effective for image-related tasks because they automatically learn important visual features such as edges, textures, and shapes (Goodfellow et al., 2016). Transfer learning was presented as a technique that allows developers to adapt pre-trained models to new classification tasks while reducing training time and data requirements.

During the workshop, I encountered several challenges. One challenge involved understanding the required image dimensions and why all images needed to be resized to a consistent size before being processed by the model. Initially, I was confused about the purpose of the input height and width values and how they affected the neural network. After reviewing the notebook and examining the data pipeline, I realized that neural networks require consistent input dimensions to process images correctly. Another challenge was interpreting tensor outputs and understanding how image data is represented numerically. Seeing large tensors filled with numbers was initially overwhelming, but I learned that these values simply represent pixel information that the model uses during training. I also faced technical difficulties when exporting the completed notebook into PDF format. After experimenting with several export methods and troubleshooting software issues, I was able to generate a readable PDF suitable for submission. These challenges reinforced the importance of persistence and problem-solving when working with machine learning projects.

One of the most valuable insights I gained from this workshop was that machine learning involves much more than simply training a model. Data preparation, validation, and evaluation are equally important components of the process. I learned that a model's performance should not be judged solely by its training results because strong training accuracy does not always guarantee good performance on unseen data. Validation accuracy provides a better measure of how well a model generalizes to new examples. I also gained a better understanding of how loss functions and optimizers work together during training. The loss function measures prediction errors, while the optimizer updates the model's parameters to reduce those errors over time (PyTorch Contributors, 2025). Observing the model improve across multiple training epochs helped me understand how neural networks gradually learn meaningful patterns from data and improve their predictions.

The techniques learned in this workshop have many important real-world applications. Image classification is widely used in healthcare to assist medical professionals in analyzing X-rays, MRIs, and other diagnostic images. It is also used in facial recognition systems, autonomous vehicles, security monitoring, manufacturing quality control, and content organization on social media platforms (Szeliski, 2022). Many companies rely on computer vision systems to automate tasks that would otherwise require significant human effort. Understanding the foundations of image classification provides valuable insight into how these technologies operate and how artificial intelligence can be applied to solve practical problems.

Overall, this workshop was a valuable learning experience that strengthened both my technical knowledge and my confidence in working with artificial intelligence tools. I enjoyed seeing how theoretical concepts from class could be applied to a real image classification problem. Successfully training a model and observing its predictions made the learning process more engaging and meaningful. The workshop also reinforced my interest in artificial intelligence and computer vision, which are important areas within my AI and Robotics degree program. Moving forward, I would like to explore convolutional neural networks and transfer learning in greater depth because of their widespread use in modern computer vision systems. This experience provided a strong foundation for future projects and helped me better appreciate the complexity and potential of machine learning technologies.

References

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